

Homework 2: Operations on Signals CPE 381

Instructor: Rahul Bhadani

Due: Sept 27, 2026, 11:59 PM
116 points

You are allowed to use a generative model-based AI tool for your assignment. However, you must submit an accompanying reflection report detailing how you used the AI tool, the specific query you made, and how it improved your understanding of the subject. You are also required to submit screenshots of your conversation with any large language model (LLM) or equivalent conversational AI, clearly showing the prompts and your login avatar. Some conversational AIs provide a way to share a conversation link, and such a link is desirable for authenticity. Failure to do so may result in actions taken in compliance with the plagiarism policy.

Additionally, you must include your thoughts on how you would approach the assignment if such a tool were not available. Failure to provide a reflection report for every assignment where an AI tool is used may result in a penalty, and subsequent actions will be taken in line with the plagiarism policy.

Submission instruction:

Upload a .pdf on Canvas with the format CPE381-LastFirst-HW-XX. For example, if your name is Sam Wells, your file name should be CPE381-WellsSam-HW-XX.pdf. If there is a programming assignment, then you should include your source code along with your PDF files in a zip file CPE381-LastFirst-HW-XX.zip. Your submission must contain your name and UAH Charger ID or your UAH email address. Please number your pages as well.

Theory

1 Linear Interpolation (6 Points)

Assume three signal values $x = \{3.0, 4.0, 3.5\}$ at timestamp $t = \{2.0, 4.0, 6.0\}$. We will consider the case of linear interpolation.

1. How many linear segments are required if given three samples or signal values? **(2 Points)**

2. Write down all the equation of line segments necessary for linear interpolation given the three samples above. **(2 Points)**
3. What is the value of when interpolated at $t = 2.5$? **(2 Points)**

2 Cubic Spline Interpolation (10 Points)

Consider signal values $x = \{0.0, 7.0, 9.0, 3.0, 4.0\}$ at timestamp $t = \{0.0, 1.0, 2.0, 4.0, 5.0\}$. We will consider the case of cubic spline interpolation.

1. Assuming free boundary condition, write the B matrix corresponding to all the b_i coefficients as per formulation provided in Chapter 02 Slides. **(6 Points)**
2. What are the values of a_1, b_1, c_1, d_1 following the convention for cubic spline interpolation formulation as described in the Chapter 02 slides? **(4 Points)**

3 Play with Signals. (10 points)

A continuous-time signal $x(t)$ is shown in Figure 1.

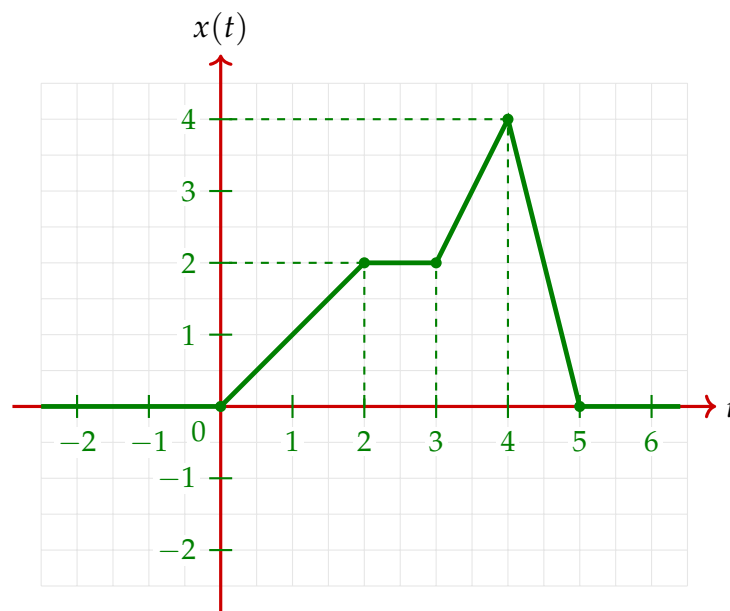


Figure 1: The ramp rises with unit slope until $t = 2$, holds the value 2 over $2 \leq t \leq 3$, climbs with slope 2 to its peak of 4 at $t = 4$, and falls to zero at $t = 5$.

Based on $x(t)$ plot, sketch and label each of the following signals: (i) $x(t - 2)$; (ii) $x(2t)$; (iii) $x(t/2)$; (iv) $x(-t)$; (v) $-x(-t)$.

4 Picasso of Signals (15 points)

Sketch the following signals:

1. $x(t) = u(t) - u(t - a)$, $a > 0$
2. $y(t) = t[u(t) - u(t - a)]$, $a > 0$
3. $z(t) = u(-t) - u(t)$ where $u(t)$ is the unit-step signal.

5 Divide and Conquer (10 points)

Determine the even and odd components of the following signals:

1. $x(t) = u(t)$.
2. $x(t) = \sin\left(\omega_0 t + \frac{\pi}{4}\right)$

6 Rinse and Repeat (15 points)

Determine whether or not each of the following signals is periodic. If signals are periodic, find their fundamental period.

1. $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$.
2. $x(t) = \cos^2(t)$.
3. $x(t) = \cos(2\pi t)u(t)$.

Practice

7 MATLAB Programming: Signal Plotting (10 points)

Consider

$$y(t) = A \cos(\Omega_c t + s(t))$$

1. $A = 1$, $\Omega_c = 2$, $s(t) = \frac{t^2}{4}$. Use MATLAB to plot the signal for $0 \leq t \leq 40$ seconds in the steps of 0.05. Use sound function to listen to the signal.
2. $A = 1$, $\Omega_c = 2$, $s(t) = -2\sin(t)$. Use MATLAB to plot the signal for $0 \leq t \leq 40$ seconds in the steps of 0.05.

8 Compact Form (20 points)

For Figure 2, represent the signal (f) in compact form using transformed version of unit-step function $u(t)$ and ramp signal $r(t)$ as necessary.

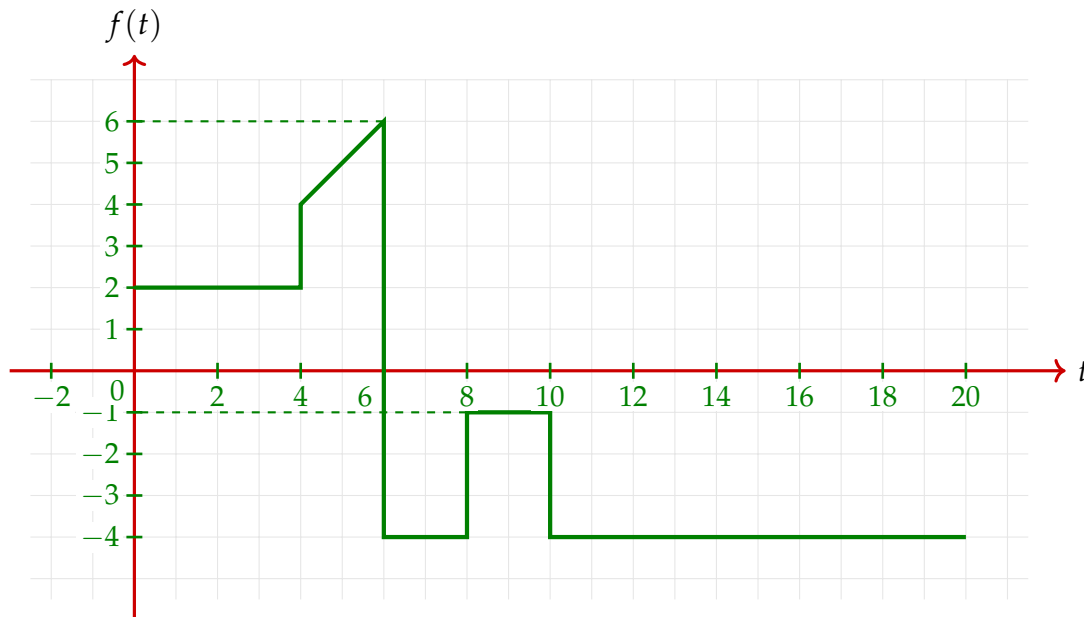


Figure 2: Compact Form

Write a MATLAB script to produce Figure 2. Note that the signal for $t < 0$ is zero.

Tip: You may use heaviside function in MATLAB to implement $u(t)$.

9 Cubic Spline Interpolation and Data-fitting (20 Points)

Consider renewable energy based electric grid and weather data for 1 week starting from 1/1/2018 12:00:00 AM in the zone 1 of ERCOT (Electric Reliability Council of Texas) that operates Texas's electrical grid, the Texas Interconnection. Data can be downloaded from https://github.com/rahulbhadani/CPE381_FA26/blob/main/Data/week_data_ERCOT_zone_1_.csv.

For those who are interest, the original source of data is <https://doi.org/10.5281/zenodo.5130611>.

The time column of the given CSV file is of the type `datetime`. `datetime` can be converted to POSIX format using

```
csv_file = 'week_data_ERCOT_zone_1_.csv';
week_data = readtable(csv_file);
px_time = posixtime(week_data.time);
```

The first entry, of the `px_time` is 1514764800.

You can subtract the entire time column from the first value to make the first time value 0:

```
px_time = px_time - px_time(1);
```

For the give time series data, perform the following task:

1. Plot a line graph of the load power from the given data against `px_time`. Provide the figure as a part of your submission as well. **(5 Points)**
2. As the given data has time-granularity of 1 minute, the sampling time $\Delta t = 60$ s, we are interested in resampling the data at $\Delta t = 1$ s.

Write a MATLAB script to resample the load power data at $\Delta t = 1$ s using cubic spline interpolation technique. Use free boundary condition. You may use the cubic spline interpolation function implementation provided as a part of the course GitHub repository. **(10 Points)**

3. Plot the original load power data and resampled load power data together. Set x-axis limit to `[7100,7300]`. Provide the figure as a part of your submission as well. **(5 Points)**

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